

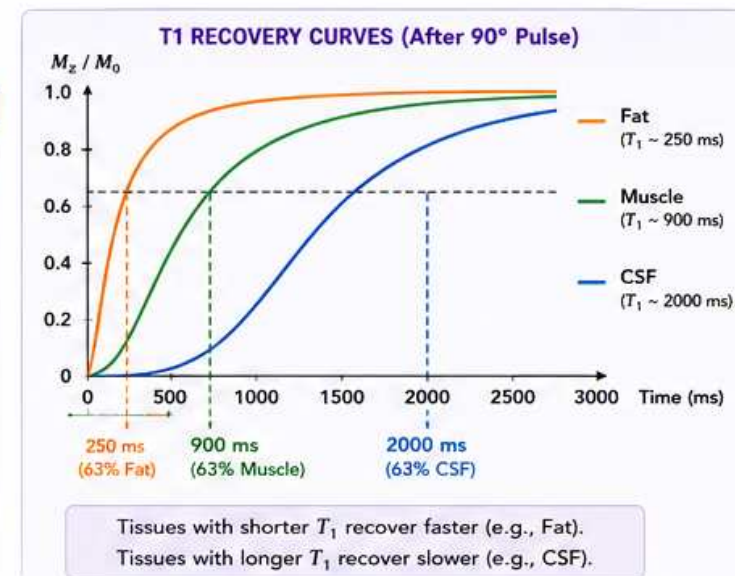
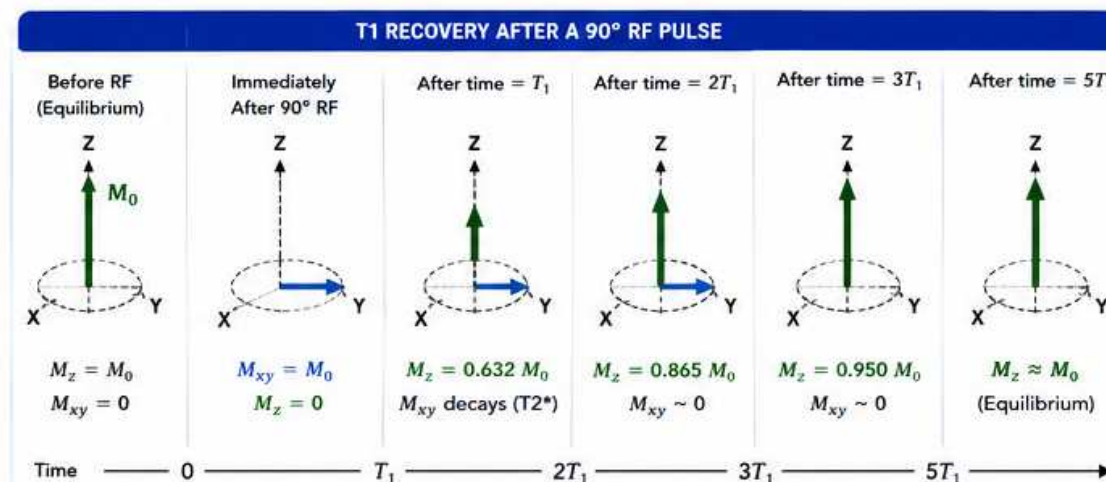
1.9 T1 RECOVERY (LONGITUDINAL RELAXATION)

After an RF pulse tips M_z away from +Z, it recovers back toward equilibrium with time constant T_1 .

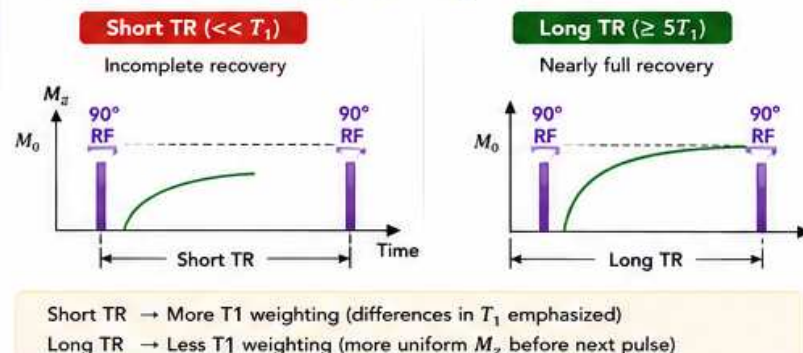
T_1 (spin-lattice relaxation) is the process by which longitudinal magnetization (M_z) returns to equilibrium (+Z) after being disturbed by an RF pulse.

Energy is exchanged between the spins and the surrounding lattice (tissue environment).

The time constant T_1 is the time it takes for M_z to recover to 63.2% of its full recovery.



EFFECT OF REPETITION TIME (TR) ON M_z RECOVERY



TYPICAL T1 VALUES AT 1.5T (approx.)

Tissue	T_1 (ms)
Fat	200 – 300
Liver	600 – 800
Skeletal Muscle	800 – 1000
Gray Matter	1000 – 1300
White Matter	600 – 900
CSF	1800 – 2500

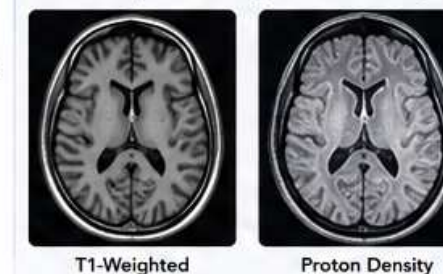
Values vary with field strength, sequence, and vendor.

CLINICAL RELEVANCE (T1 WEIGHTING)

With short TR and short TE:

- ✓ Tissues with short T_1 (fat) have more M_z → produce stronger signal → appear bright
- ✓ Tissues with long T_1 (CSF) have less M_z → weaker signal → appear dark
- ✓ Used in SE, IR, and many 3D sequences (e.g., T1-weighted FSPGR, MPRAGE)

Example: T1-Weighted Brain Image



In T1-weighted images: Fat (bright) > Gray Matter > White Matter > CSF (dark)

ENGINEERING INSIGHT

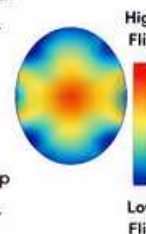
- Accurate T_1 contrast depends on:
 - Correct TR selection
 - Consistent flip angle
 - Stable B_0 (frequency) and B_1 (flip angle)
- M_z recovery occurs in the rotating frame off, between RF pulses.



FIELD SERVICE TIP

- Verify sequence TR is as prescribed. Wrong TR → unexpected contrast.
- Check flip angle calibration (affects how much M_z is tipped and how much can recover).
- Inhomogeneous B_1 → variable flip angle → inconsistent T_1 contrast.

B_1 Field Map (Example)



COMMON MISCONCEPTION

“Long TR always gives better T_1 contrast.”

Reality: Very long TR reduces scan efficiency (longer scan time) and SNR per unit time. Optimal TR balances contrast, SNR, and scan efficiency.

REAL-WORLD ANALOGY

Think of a spring pulled down (energy removed) and then released. It returns to its rest position with a time constant (T_1). Different tissues are like springs with different stiffness (recovery speed).



KEY TAKEAWAY

T_1 recovery is the return of M_z to +Z (equilibrium) after RF excitation.

The time constant T_1 describes how fast this recovery happens for each tissue.

Proper TR selection is the key to controlling T_1 contrast and image appearance.

